Continued fractions

Presenter notes for two 50-minute lectures

The slides follow Chapter 3 of *Patterns, Proofs, and Primes*. Each lecture has its own 50-minute route. The suggested times include student calculation and discussion. Answer slides follow question slides, so advancing the PDF provides a simple reveal without animation software. The optional slides come after the exit question.

Before the first lecture. Students should know how to multiply and invert a 2×2 matrix, how a simple continued fraction is written, and what its convergents are. Lecture 1 reviews the transformation rules. For Lecture 2, recall that the convergents tend to the original irrational number and their denominators are eventually unbounded. No calculus or abstract algebra is needed for these proofs.

Using the files. Share only the slide PDF window in Zoom and keep this notes document separate. If you have one screen, practice switching to the notes without changing the shared window. The editable LaTeX sources are included in the source bundle. No live web services or fonts need to download during the lectures.

Pacing. Lecture 1 reaches the prefix-conjugation identity at minute 29 and finishes that argument by minute 35. Lecture 2 starts with the converse of Lagrange's theorem and a direct calculation for $\sqrt{3}$. Slides 6–8 prove the conjugation identities. The integer-pair proof concludes at minute 34; the $\sqrt{6}$ exercise and the Lagrange checkpoint take us to minute 39. Galois begins only then, and its proof ends at minute 47. The main route still totals 50 minutes. If the class needs more time on Lagrange, use the last eleven minutes to revisit the conjugate bounds and the exercise, and start the next meeting with slide 24, on pure periodicity. Keep the student calculation break whenever possible.

Lecture 1: Continued fractions and matrix fixed points

1. Continued fractions and matrix fixed points

Minutes 0 to 1

Open with the question: why should repeating continued fractions know anything about quadratic equations? Students have already met linear fractional transformations. Treat the first few slides as a review, not a new chapter of prerequisites. Today's destination is the prefix-conjugation argument. Source: course book, Sections 3.2 and 3.7.

2. A repeating pattern

Minutes 1 to 3

Ask students which digit belongs to the repeating block in each expression. Explain that pure periodicity includes the integer part. A bar over the block means to repeat it indefinitely. The original number and its complete quotient can have different kinds of periodicity. Recall that rational simple continued fractions terminate. In these lectures, a finite rational expansion does not count as eventually periodic. Source: course book, Section 3.7.

3. The two periodicity theorems

Minutes 3 to 5

State the destination, without trying to prove both results now. Use $\sqrt{2}$ as the simplest example of a quadratic irrational. Today proves the forward direction of Lagrange. Lecture 2 proves the converse and both directions of Galois. Clarify the names gently: Legendre's continued-fraction theorem concerns exceptionally good rational approximations. Lagrange is the eventual-periodicity theorem. The theorem statements use simple continued fractions. Source: course book, Sections 3.7–3.9. External cross-check: Corless, Periodic and Fixed Points of the Gauss Map.

4. A Möbius transformation

Minutes 5 to 7

Use the notation from your preceding lecture, while making clear that $T_M(t)$ is a fractional action rather than ordinary matrix multiplication by a scalar. The denominator condition is essential. With rational coefficients, any finite pole is rational, so an irrational input never hits a pole. Infinity is a convenient optional convention, not a prerequisite for today's proofs. Source: book 3.2.

5. Why a matrix acts on a number

Minutes 7 to 9

This is the projective viewpoint without projective-space terminology. Emphasize the ratio. The columns are ordinary vectors, and the fractional transformation records their coordinate ratio. Ask why M and $-M$ produce the same map. The identity matrix and all its nonzero scalar multiples therefore represent the identity transformation. We will need that observation when excluding degenerate fixed-point equations. Source: book 3.2 and Appendix A.2.

6. Composition

Minutes 9 to 12

Here M is translation by 1 and N is inversion. Compute the product with the class, then compare it with the nested function. If students already remember the general substitution proof, do not repeat its four coefficients. The first backup slide contains that calculation if requested. Ask what happens in the opposite order: $NM = \begin{pmatrix} 0 & 1 \\ 1 & 1 \end{pmatrix}$ acts by $1/(t+1)$. This makes the order of multiplication memorable. Source: book 3.2.

7. An inverse, by solving for t

Minutes 12 to 14

Let students supply the rearrangement $(a - cy)t = dy - b$. An important corollary is that an invertible rational Möbius transformation preserves irrationality: if its output were rational, its inverse would make the input rational. There is no hidden pole in this argument. For $y = T_M(t)$, $a - cy = (ad - bc)/(ct + d) \neq 0$. We use this fact later to rule out a merely linear equation. Source: book 3.2 and the proof in 3.7.

8. One continued-fraction digit

Minutes 14 to 16

The tail t here is a whole remaining continued fraction. The letter D in $D(a)$ means a digit matrix. In Lecture 2, the plain letter D will instead denote a discriminant under a square root; point out the contextual distinction if needed. All finite digit products have determinant ± 1 and integer inverses. This is the exact sense in which we can undo a prefix. Source: book 3.2–3.3.

9. A block of digits

Minutes 16 to 18

Translate the displayed continued fraction back into nested functions. All digits after the integer part are positive. The last entry t is the tail, so it need not be an integer. The example fixes the period matrix that will run through the rest of the lecture. Have students keep $\begin{pmatrix} 3 & 1 \\ 2 & 1 \end{pmatrix}$ on their paper. Source: book 3.2 and the worked example in 3.7.

10. A repeating tail

Minutes 18 to 20

This is the central idea. Pause before showing the fixed-point equation and ask what remains after removing a complete block. An infinite simple continued fraction has a well-defined irrational limit; this follows from the convergent bounds already developed in class. A backup gives the short irrationality argument. Positivity of the period digits makes B, C strictly positive, even for a one-digit period. Source: book 3.7.

11. A fixed point gives an equation

Minutes 20 to 22

Keep this algebraic step brief. The nontrivial part will be understanding exactly which quadratics repeat. At this point the number is irrational because the continued fraction is infinite, so it is a quadratic irrational, not merely some root of a reducible quadratic. Contrast the period matrix with the identity: $t = T_I(t)$ only yields $0 = 0$. Source: book 3.7.

12. Your turn: the block $1, 2$

Minutes 22 to 25

Give students a real minute of silence, then ask for a quadratic before asking for its roots. Expected equation: $2x^2 - 2x - 1 = 0$. Expected roots: $(1 \pm \sqrt{3})/2$. The continued fraction exceeds 1, so it is the positive root. If someone takes the negative root, ask which interval contains $1 + 1/(2 + 1/(\cdots))$. Advance only after discussion. This is book Exercise 15, expanded into a classroom prompt.

13. The periodic root

Minutes 25 to 27

The prime denotes the other algebraic root, not a derivative. Do not yet explain the significance of the interval $(-1, 0)$. Point out the pattern and promise that Lecture 2 will explain it exactly. One can check the value by substituting it into $(3x + 1)/(2x + 1)$. Source: book 3.7.

14. A finite prefix

Minutes 27 to 29

Emphasize the distinction between a periodic tail and the original number. We know a quadratic equation for the tail x, but must produce one for the number y with its prefix. Let students suggest substitution or inversion. Direct substitution works, but conjugation packages it into one reusable identity. Source: book 3.7, equation (3.3).

15. The prefix conjugation identity

Minutes 29 to 32

Read the composition from right to left: remove the prefix, apply the period to its fixed tail, and put the prefix back. Spend the full three minutes on why each equality holds. For students comfortable with function notation, write MNM^{-1} directly while saying that the matrices act fractionally. An empty prefix means $M = I$. Source: book 3.7, equation (3.3).

16. Why the new equation is genuinely quadratic

Minutes 32 to 35

This is the rigor checkpoint. If $K = \lambda I$, then $N = M^{-1}KM = \lambda I$, contradicting the positive lower-left entry of the period matrix. Thus the three coefficients $c, d - a, -b$ are not all zero. If $c = 0$, a nonzero rational linear polynomial would vanish at the irrational y, which is impossible. Rationality of a Möbius image would also imply rationality of its inverse image. The backup fixed-point lemma gives this argument in isolation. Source: book 3.7.

17. The equation for $\sqrt{3}$

Minutes 35 to 38

Have students check at least one row of the matrix product. For an intermediate check, $MN = \begin{pmatrix} 5 & 2 \\ 3 & 1 \end{pmatrix}$. The positive sign follows from $y = 1 + 1/x > 0$. This example both verifies the familiar continued fraction of $\sqrt{3}$ and shows what conjugation actually does to the quadratic equation. Source: book 3.7 and Exercise 21.

18. Your turn: a different prefix

Minutes 38 to 41

Allow about 90 seconds for the calculation. Students who know $1/x = \sqrt{3} - 1$ can immediately identify $y = 1+\sqrt{3}$, but ask them to check the conjugated matrix too. The result is $\begin{pmatrix} 3 & 2 \\ 1 & 1 \end{pmatrix}$, giving $y^2 - 2y - 2 = 0$. The digit stream is $2, 1, 2, 1, \ldots$, so its pure period is $[\overline{2, 1}]$. Source: book Exercise 21.

19. A prefix that disappears

Minutes 41 to 43

Distinguish a particular written decomposition into prefix and period from the shortest possible preperiod of the actual digit sequence. Here the first digit already belongs to a cycle of length two. This is an important example to bring back after Galois's criterion, since the conjugate $1 - \sqrt{3}$ lies in $(-1, 0)$. Source: book 3.9.

20. Quadratic conjugates

Minutes 43 to 45

Assume D is a positive nonsquare integer. Clarify again that the prime is algebraic conjugation, not differentiation. All we need is that changing the sign of $\sqrt{D}$ commutes with adding, multiplying, and taking reciprocals when the denominators are nonzero. The complete quotients are rational transformations of the original surd, so their conjugates follow the same algebra. Source: book 3.7–3.8.

21. Reduced surds

Minutes 45 to 48

Ask which inequality fails in each of the two negative examples. For $\sqrt{2}$ it is the lower bound on the conjugate. For $\sqrt{2}/2$ it is $x > 1$. Galois says that this definition is exactly the right one for pure periodicity. Today students need the definition and examples; Lecture 2 explains the mechanism. Source: book 3.8–3.9.

22. What we have proved

Minutes 48 to 50

Expected exit answer: the identity transformation fixes every real number, so the resulting polynomial is $0 = 0$. We needed a nonscalar transformation, supplied by the positive lower-left entry of the period matrix. Stop here at 50 minutes. The remaining slides are backups, not part of the planned lecture. Suggested preparation: review complete quotients and the fact that convergents p_k/q_k tend to the original irrational.

Optional slides after the 50-minute stopping point

Backup 1. Composition in coordinates

Use only if students request the general composition calculation. The equation holds wherever the displayed finite formulas are defined, and globally if infinity is included in the projective convention. Source: book 3.2.

Backup 2. The nonscalar fixed-point lemma

This separates the elementary lemma from any special continued-fraction properties. Note that being invertible does not by itself exclude the identity transformation. One needs the nonscalar assumption as well as an irrational fixed point. Source: book Lemma 3.12.

Backup 3. Why an infinite simple continued fraction is irrational

The alternating-convergent argument produces a limit for any infinite positive digit sequence, not just for the expansion of a number already known to be irrational. For a rational candidate p/q in lowest terms, increasing denominators ensure that the convergents eventually differ from it. The integer numerator of their difference then has absolute value at least one. A weak upper bound is enough here. Source: book 3.7.

Lecture 2: Why quadratic surds have repeating continued fractions

1. Why quadratic surds have repeating continued fractions

Minutes 0 to 1

Begin with the question left open by Lecture 1. We know how a repeating continued fraction produces a quadratic equation. Today we start with a quadratic surd and prove that its continued fraction must eventually repeat. Establish that result before introducing Galois's additional criterion. The proof uses rationalization, convergence of convergents, and the pigeonhole principle. Source: book 3.8–3.9.

2. The converse we still need

Minutes 1 to 2

The two implications together are usually called Lagrange's theorem. Here *the converse* means the direction left for this lecture. A quadratic surd is an irrational real root of an integer quadratic; we will spell out its square-root form shortly. An overline means that the indicated block repeats forever. The theorem makes no claim yet about where the repetition begins. All continued fractions here are simple; rational finite expansions are not called eventually periodic. Source: book Theorem 3.13.

3. The algorithm and its remaining tails

Minutes 2 to 4

Explain why the strict fractional-part inequalities continue to hold. If a later complete quotient were rational, then $x_n = a_n + 1/x_{n+1}$ would make every preceding one rational. Thus the algorithm never terminates for our irrational input. Rebuild the index convention: x_0 is the original number, a_0 its integer part, and x_1 the first remaining tail. Source: book 3.5 and 3.8.

4. First see repetition happen: $\sqrt{3}$

Minutes 4 to 6

Lecture 1 started from repeating digits and identified their value. Here the direction is reversed: start with $\sqrt{3}$ and run the algorithm until an exact tail returns. Use $1 < \sqrt{3} < 2$ to check the two later integer parts. In the middle line, $2/(\sqrt{3}-1) = \sqrt{3}+1$. Stress that the certificate is equality of whole tails, not merely the occurrence of the same digit twice. Source: book Example 3.16.

5. What will force a tail to return?

Minutes 6 to 7

Give the proof's purpose before introducing its algebra. Keeping the same square root is not enough: there are infinitely many integer pairs. We need both integrality and bounds. The word *state* below will mean just the pair (P_n, Q_n). The next slides establish the elementary facts about conjugates that this proof needs, including why a conjugate follows the same rational formulas. Source: book 3.8.

6. A surd and its other root

Minutes 7 to 9

This proves the connection between the square-root expression and the other root; it is not just a rule to memorize. Clear denominators in the displayed polynomial to get integer coefficients. Conversely, the quadratic formula puts every real quadratic irrational in this form. Its discriminant is positive and nonsquare. The representation in a fixed square root is unique: equality of two different rational coefficients of $\sqrt{D}$ would make $\sqrt{D}$ rational. Any rational quadratic having x as a root is a scalar multiple of this one: cancel their quadratic terms, and a remaining nonzero linear polynomial could not vanish at the irrational x. Thus the other root is unambiguous. The prime is not a derivative. Source: book 3.7.

7. Why conjugation respects arithmetic

Minutes 9 to 11

Extend the sign-change operation to all numbers in the same square-root field, including rational ones with square-root coefficient zero. No field terminology is needed on the slide. The multiplication identity works because $(\sqrt{D})^2 = D$ and $(-\sqrt{D})^2 = D$; the rational term is unchanged and the square-root term changes sign. This is the explicit justification for the arithmetic assertion at the end of Lecture 1. It concerns algebraic conjugation of numbers; the matrix operation MNM^{-1} from Lecture 1 is a different use of the word. Source: book 3.7, with the arithmetic expanded.

8. Why conjugation respects reciprocals

Minutes 11 to 13

Justify nonvanishing, rather than silently dividing. If $z' = u - v\sqrt{D} = 0$ and $v \neq 0$, then $\sqrt{D} = u/v$ would be rational. If $v = 0$, then $z' = u = z \neq 0$. Thus $z' \neq 0$ whenever $z \neq 0$, and zz' is nonzero. The last line assumes rational a, b, c, d and a nonzero denominator. Its conjugate denominator is nonzero by the same argument. These two arithmetic slides prove, in particular, $(T_M(x))' = T_M(x')$ for Lecture 1's rational matrices. Source: book 3.7–3.8, with the rationalization expanded.

9. Step 1: start with an integer pair

Minutes 13 to 15

This shows existence of the integer representation, including the divisibility condition that will make the recurrence integral. If the desired root initially has a minus sign before the square root, replace (A, B, C) by $(-A, -B, -C)$: $(B + \sqrt{D})/(-2A) = (-B - \sqrt{D})/(2A)$. The discriminant does not change. Since x is a real irrational, $D > 0$ and D cannot be a square. There is no need to make D squarefree. For special examples we can choose a smaller D directly: $\sqrt{3} = (0 + \sqrt{3})/1$ has the required divisibility. Source: book 3.8, Step 2.

10. Derive the rule by rationalizing

Minutes 15 to 18

Start by subtracting the digit explicitly: $x_n - a_n = (P_n + \sqrt{D} - a_n Q_n)/Q_n = (\sqrt{D} - P_{n+1})/Q_n$. Taking the reciprocal gives the first displayed line. Multiply numerator and denominator by $\sqrt{D} + P_{n+1}$ to get the second. This derives both formulas and explains the sign in P_{n+1}. The new denominator is not yet known to be an integer; the next slide proves exactly that. The square root and D stay fixed throughout. Source: book equation (3.4).

11. Why the new pair is still integral

Minutes 18 to 20

This is an induction with a base case on the initialization slide. The recurrence makes P_{n+1} an integer immediately. To see the divisibility for Q_{n+1}, subtract $P_{n+1}^2 - P_n^2$ from $D - P_n^2$; both are multiples of Q_n. The final equality establishes the next induction hypothesis. Divisibility is valid with negative denominators as well, so no positivity assumption has slipped into Step 1. Source: book 3.8, Step 2.

12. Step 2: can these integers keep growing?

Minutes 20 to 21

Pause at the gap in the argument. The same square root has persisted, but that fact alone does not force any repetition. The new ingredient is an inequality for the conjugate. Avoid introducing Galois's theorem or the name *reduced* here: the inequalities are being proved as tools for Lagrange's converse. Source: book 3.8.

13. The conjugate follows the same algebraic step

Minutes 21 to 23

Every equality on the first display follows from the previous arithmetic slides: reciprocals commute with conjugation, and the integer a_n remains fixed. This sequence of conjugates is usually not the ordinary continued-fraction algorithm started at x'. The example makes the difference concrete. Each x'_n is irrational and hence cannot equal the integer a_n, so every displayed denominator is nonzero. Source: book 3.8, Step 1.

14. Express a tail using the original number

Minutes 23 to 25

Recall the prefix matrix explicitly: $D(a_0) \cdots D(a_{n-1}) = \left(\begin{smallmatrix} p_{n-1} & p_{n-2} \\ q_{n-1} & q_{n-2} \end{smallmatrix}\right)$. It acts on the unprocessed tail x_n to give x. The displayed rearrangement derives the inverse-prefix identity instead of citing it without calculation. The denominator cannot vanish: x is irrational whereas p_{n-1}/q_{n-1} is rational. For $n \geq 2$, both relevant q's are positive. Source: book 3.2 and 3.8.

15. Why the conjugates eventually become negative

Minutes 25 to 28

Spend time on the sign argument. The convergent lies within half the distance from x to x', so it is on the same side of x' as x. Equivalently, $x' - p_k/q_k = (x' - x) + (x - p_k/q_k)$, and the second term has magnitude less than half that of the nonzero first term. The difference is therefore positive if $x' > x$ and negative if $x' < x$. For large n this applies at both indices $n-2, n-1$. The two positive denominators q_{n-2}, q_{n-1} preserve the signs, and the minus sign in front of the ratio makes x'_n negative. Only convergence has been used, with no periodicity assumption. The argument covers $x < 0$ and $x' > x$ as well. Source: book 3.8, Step 1.

16. One more step gives a fixed interval

Minutes 28 to 29

The interval follows from an elementary reciprocal inequality: if $t < -1$, then $-1 < 1/t < 0$. Negativity was proved on the previous slide; this slide proves entry into the smaller interval and that later conjugates stay there. These are separate facts. The restriction $n \geq 1$ ensures $a_n \geq 1$, even if the original number has a negative integer part. Source: book 3.8, Step 1.

17. Bound the denominator first

Minutes 29 to 30

Keep the sign check visible. We are not assuming that the initial denominator was positive: the eventual inequalities force positivity here. Only after that check is it legitimate to multiply an inequality by Q_n without reversing its sign. The upper bound depends solely on the fixed integer D. Source: book 3.8, equation (3.5).

18. Now bound the numerator integer

Minutes 30 to 31

This is the second half of the integer bounds. Every inequality now appears on the slide: multiply by a positive Q_n, add $\sqrt{D}$, and substitute the already proved upper bound on Q_n. Students should see why both the original number and its conjugate were useful. One bounded real expression alone would not have bounded both integers. Source: book 3.8, equation (3.5).

19. Step 3: only finitely many pairs remain

Minutes 31 to 33

Invoke the pigeonhole principle explicitly. For $D = 3$ the box contains at most nine integer pairs; some do not satisfy all the original conditions, which only makes the possible set smaller. We do not need to enumerate the actual orbit. For general D, there are at most $(2\lfloor\sqrt{D}\rfloor + 1)\lfloor 2\sqrt{D}\rfloor$ pairs in the box. Finiteness is all the proof needs. The first repeated pair need not involve index zero. Source: book 3.8, Step 3.

20. A repeated pair forces repeating digits

Minutes 33 to 34

This is the precise link from pigeonhole to continued fractions. The update rule is deterministic: equal complete quotients produce equal digits and equal successors. Induction gives the entire repeated future. The displayed period has positive length $j - i$, although it need not be the shortest period. The prefix is omitted if $i = 0$. At minute 34 the main existence theorem is fully proved, without using Galois's theorem. Source: book 3.8.

21. Your turn: find the period of $\sqrt{6}$

Minutes 34 to 37

Allow two minutes of calculation and one minute for comparison. The first digit is 2; the states are $(0, 1)$, $(2, 2)$, $(2, 1)$, $(2, 2)$, with digits $2, 2, 4, 2$. If a student stops when a digit repeats, point to the requirement that the entire pair repeat. If a denominator is unexpectedly negative here, check the sign in P_{n+1}. No decimal approximation is needed: $2 < \sqrt{6} < 3$ determines every digit in this example. Source: book equation (3.4).

22. The calculation for $\sqrt{6}$

Minutes 37 to 38

Check the first transition explicitly: $P_1 = 2 \cdot 1 - 0 = 2$, $Q_1 = (6 - 4)/1 = 2$. The next pair is $(2, 1)$, then $(2, 2)$ returns. Although $a_0 = a_1 = 2$, their next digits differ: $a_1 = 2$ and $a_2 = 4$. Equal digits do not determine the next tail. The equality at indices 1 and 3 is the certificate. The two cyclic tails have conjugates in $(-1, 0)$, just as the proof predicts.

23. Lagrange's theorem is now complete

Minutes 38 to 39

This is a natural checkpoint at minute 39, after proof and practice. Ask a student to recall what the conjugate contributed: it supplied the inequalities that bounded the two integers. Only now move to Galois's additional assertion. If the class needs more time, use the final eleven minutes to revisit the bounds and the exercise, and begin the next meeting with the following slide. The complete Lagrange proof stands alone. Source: book Theorem 3.13.

24. Now ask for pure periodicity

Minutes 39 to 41

Define the name using the inequalities already established. Galois adds an exact criterion for removing the initial segment; Lagrange has already established existence of a period for every quadratic surd. In the example, $-\sqrt{3} < -1$, whereas $(1 - \sqrt{3})/2$ is between -1 and 0 and its partner exceeds 1. We will first prove sufficiency, then necessity. A reduced x_0 has $a_0 \geq 1$, so the invariance argument applies from the very first step as well. Source: book Theorem 3.17.

25. Finiteness alone permits an initial segment

Minutes 41 to 42

This isolates the additional logical task. Knowing that reduced tails stay reduced and lie in a finite set does not yet imply pure periodicity. The example is a finite forward-invariant system with a transient first state. The next slide proves uniqueness of a reduced predecessor; existence of an arbitrary predecessor is not needed for rewinding the actual orbit. Source: book 3.9.

26. Recover the previous digit from the conjugate

Minutes 42 to 45

Give the inequality its full justification: add the integer a to $0 < -u' < 1$. The unique integer part of $-1/z'$ is then a. Once a is known, the original equation recovers u. This proves at most one reduced predecessor, not uniqueness among all possible numbers preceding the same tail. Equal values of z have equal conjugates by the well-defined sign-change operation proved earlier. For a numerical check using exact surds, let $z = (1+\sqrt{3})/2$. Then $-1/z' = 1+\sqrt{3}$, so the previous digit is 2 and $u = 2+1/z = 1+\sqrt{3}$. Source: book equation (3.6).

27. Rewind the repetition to the start

Minutes 45 to 46

If $i = 0$, we already have the desired conclusion. Otherwise apply uniqueness at indices i, j, then $i-1, j-1$, and continue for exactly i steps. Each actual predecessor is reduced because the starting point was reduced and the condition persists. The index difference stays $j - i$ throughout. Forward iteration from $x_0 = x_{j-i}$ gives equality of all corresponding digits, including a_0. This proves the sufficient direction of Galois. Source: book 3.9.

28. A pure period must also be reduced

Minutes 46 to 47

The convergence argument in Lagrange's proof supplies eventual entry into the inequalities independently of Galois. Pure periodicity repeatedly returns to the original value, so that value must itself have the eventual inequalities. This proves necessity without another matrix calculation. The proof of Galois is complete at minute 47. Source: book 3.9.

29. The distinction in two examples

Minutes 47 to 49

Pause before reading the last column and invite the class to check the inequalities. Both original numbers exceed 1; it is their conjugates that distinguish them. Lagrange gives eventual periodicity in both cases, and Galois decides whether the expansion repeats from the start. The optional four-number exercise includes an example below 1 to test the other inequality. Source: book 3.9.

30. Exit question: what does each step prove?

Minutes 49 to 50

Expected answers: bounded integer pairs form a finite set, so some complete quotient and then the whole future repeat. Bounds on real numbers alone do not give a finite set, and integer pairs without bounds can escape forever. Unique reduced predecessors let us rewind a repeated state to index zero. The number $\sqrt{3}$ is a quadratic surd, so Lagrange applies; its later tails become reduced even though it is not reduced itself. Its expansion is eventually, not purely, periodic. Stop here at 50 minutes. Suggested follow-up: compute the period of $\sqrt{13}$ or use the optional reducedness exercise.

Optional slides after the 50-minute stopping point

Backup 1. Why the initial denominator can be negative

This expands the sign choice in the initialization slide. Neither D nor its nonsquare property changes when all coefficients change sign. The numerator can always carry $+\sqrt{D}$ if the denominator is allowed either sign. For this example, the first digit is -2, and the recurrence gives $(P_1, Q_1) = (4, 4)$, representing $(2 + \sqrt{2})/2$. Its conjugate is positive; eventual entry into $(-1, 0)$ is a theorem, not an assumption about the first step. Source: book 3.8, Step 2.

Backup 2. The integer pairs for $\sqrt{3}$

This revisits the opening calculation using the integer recurrence. From $(P, Q, a) = (0, 1, 1)$ we get $P_{\text{new}} = 1$ and $Q_{\text{new}} = 2$. From $(1, 2, 1)$ we get $(1, 1)$, and from $(1, 1, 2)$ we return to $(1, 2)$. The original pair happens to lie in the coarse finite box as well, but the original tail is not reduced. The box is a necessary bound for reduced states, not an equivalent definition. Source: book Example 3.16.

Backup 3. Your turn: check both inequalities

Give about 90 seconds for the comparisons, then ask for reasons. The middle two are reduced. For $\sqrt{2}$ the conjugate is below -1. For $\sqrt{2}/2$ the original number is below 1. All four are quadratic irrationals, so Lagrange gives eventual periodicity for all four. Keep the next slide hidden until the discussion finishes. Source: book 3.9 and Exercise 23.

Backup 4. The four tests, checked

For the golden ratio, use $2 < \sqrt{5} < 3$ to see that its conjugate lies in $(-1, 0)$ and the number itself exceeds 1. Its period is the digit 1. The period of $1 + \sqrt{3}$ is $2, 1$. The last example shows why the condition on the original number cannot be dropped even when the conjugate is in the right interval. Source: book 3.9.

Backup 5. A visible prefix can belong to the period

Return to the different-prefix example from Lecture 1. The digit stream is $2, 1, 2, 1, \ldots$, so the first digit already belongs to the period. A displayed prefix need not be a shortest preperiod. The earlier matrix calculation established the value; Galois now explains its classification. Source: book 3.9.

Backup 6. The conjugate reverses the period

Both B and C are positive. The transformed quadratic therefore has one positive root and one negative root. By Galois, $-1/x' > 1$; the reversed periodic continued fraction is positive too, so they are the same root. The substitution gives $C - (D_0 - A)z - Bz^2 = 0$ after multiplying by z^2; multiply by -1 to get the displayed equation. For a check, $\overline{[1, 2, 3]} = (4 + \sqrt{37})/7$ and the reversed period is $(4 + \sqrt{37})/3$. Source: book Corollary 3.19 and Exercise 26.

Backup 7. The same matrix produces Pell solutions

This optional connection is not needed for either periodicity proof. The determinant-one matrix acts both as a fractional transformation fixing $\sqrt{3}$ and as a linear transformation preserving the quadratic form $p^2 - 3q^2$. Expanding the two squares gives $4p^2 + 12pq + 9q^2 - 3p^2 - 12pq - 12q^2 = p^2 - 3q^2$. This generates infinitely many solutions, but does not by itself classify all Pell solutions. Source: book 3.10 and Exercise 18.

Sources and notation

The primary source for these lectures is the expanded course book: Chapter 3, especially Sections 3.2 and 3.7–3.10, together with Appendix A. The printed book numbers its three central identities as (3.3), prefix conjugation; (3.4), the quadratic-state recurrence; and (3.6), the unique reduced predecessor.

The notation $T_M(t)$ means the fractional action of a matrix, not ordinary multiplication by a scalar. The prime on a quadratic irrational denotes its algebraic conjugate, not a derivative. Every continued fraction in the lectures is simple, with positive digits after its integer part.

Robert M. Corless, *Continued Fractions and Chaos*, Periodic and Fixed Points of the Gauss Map, states the Lagrange and Galois characterizations. Giovanni Panti, A general Lagrange Theorem (2007), gives a broader setting for periodicity results beyond ordinary simple continued fractions. The lectures need none of those generalizations.

For a presentation setup reference, Zoom documents sharing a slide-show window. The lecture PDFs themselves do not depend on Zoom.